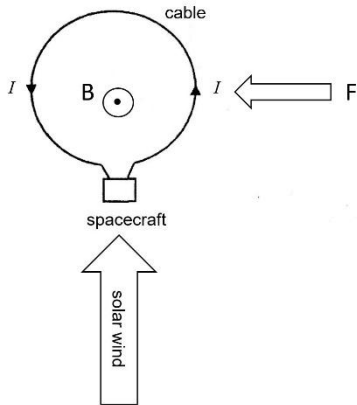


5	(a)	The current in the magsail sets up a magnetic field (which is out of the page). A charge moving in a magnetic field experiences a force, thus getting deflected. By Newton's third law, there is an equal and opposite force by the deflected charges on the magsail, thus propelling it.		B1 B1
	(b)	Photons do not have charge.		B1
	(c)	(i)		
		(ii)	$EK = \frac{1}{2} mv^2$ $500 \times 10^3 \times 1.60 \times 10^{-19} = \frac{1}{2} (1.67 \times 10^{-27}) v^2$ $v = 9.79 \times 10^6 \text{ m s}^{-1}$	C1 A1
		(iii)	$B = \frac{\mu_0 NI}{2r}$ $B = \frac{4\pi \times 10^{-7} (1) 3000}{2(64)}$ $= 29.5 \mu T$	C1 A1
		(iv)	$F = Bqv = 29.5 \times 10^{-6} \times (1.6 \times 10^{-19}) \times (9.79 \times 10^6)$ $= 4.62 \times 10^{-17} \text{ N}$	C1 A1